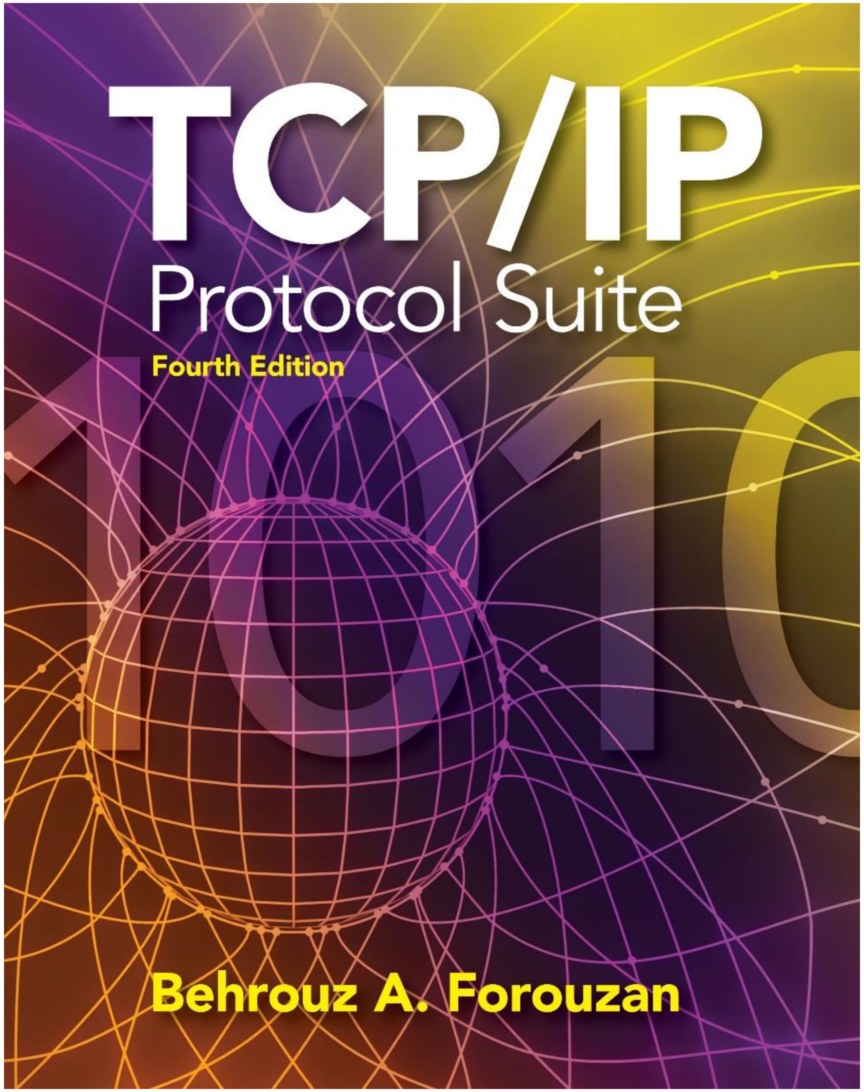
Chapter 9

Internet Control Message Protocol Version 4 (ICMPv4)



OBJECTIVES:

- To discuss the rationale for the existence of ICMP.
- To show how ICMP messages are divided into two categories: error reporting and query messages.
- To discuss the purpose and format of error-reporting messages.
- To discuss the purpose and format of query messages.
- To show how the checksum is calculated for an ICMP message.
- To show how debugging tools using the ICMP protocol.
- To show how a simple software package that implements ICMP is organized.

Chapter Outline

9-1 INTRODUCTION

The IP protocol has no error-reporting or error correcting mechanism. What happens if something goes wrong? What happens if a router must discard a datagram because it cannot find a router to the final destination, or because the time-to-live field has a zero value? These are examples of situations where an error has occurred and the IP protocol has no built-in mechanism to notify the original host.

Topics Discussed in the Section

- ✓ **The position of ICMP in the TCP/IP suite**
- ✓ **Encapsulation of ICMP Packets**

Figure 9.1 *Position of ICMP in the network layer*

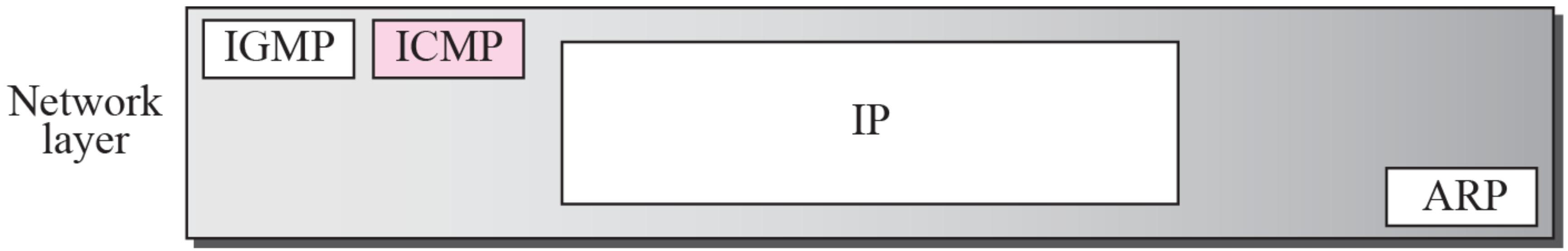
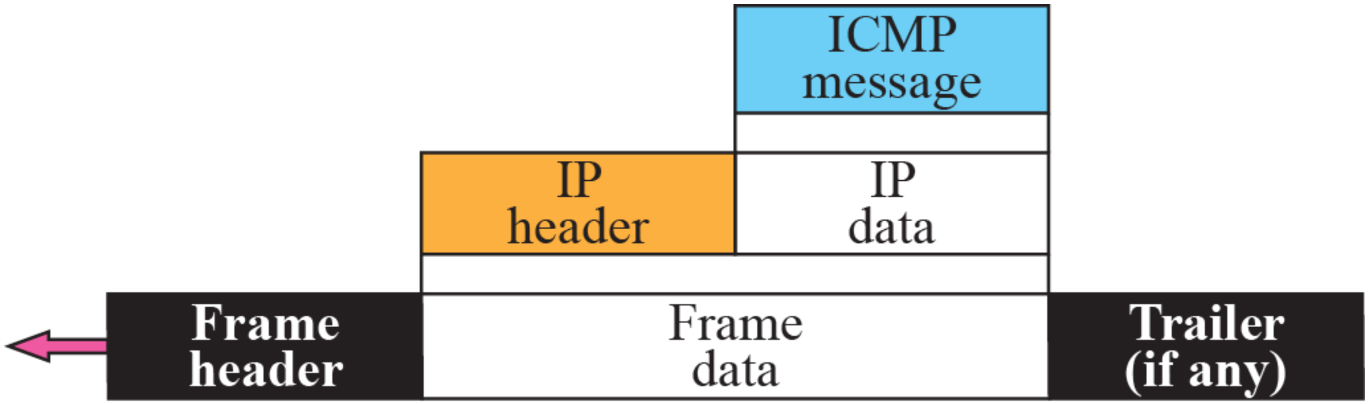


Figure 9.2 *ICMP encapsulation*



9-2 MESSAGES

ICMP messages are divided into two broad categories: error-reporting messages and query messages. The error-reporting messages report problems that a router or a host (destination) may encounter when it processes an IP packet. The query messages, which occur in pairs, help a host or a network manager get specific information from a router or another host. Also, hosts can discover and learn about routers on their network and routers can help a node redirect its messages.

Topics Discussed in the Section

- ✓ Message Format
- ✓ Error Reporting Messages
- ✓ Query Messages
- ✓ Checksum

Table 9.1 *ICMP messages*

Category	*Type*	*Message*
Error-reporting messages	3	Destination unreachable
	4	Source quench
	11	Time exceeded
	12	Parameter problem
	5	Redirection
Query messages	8 or 0	Echo request or reply
	13 or 14	Timestamp request or reply

ICMP Parameter Message Format

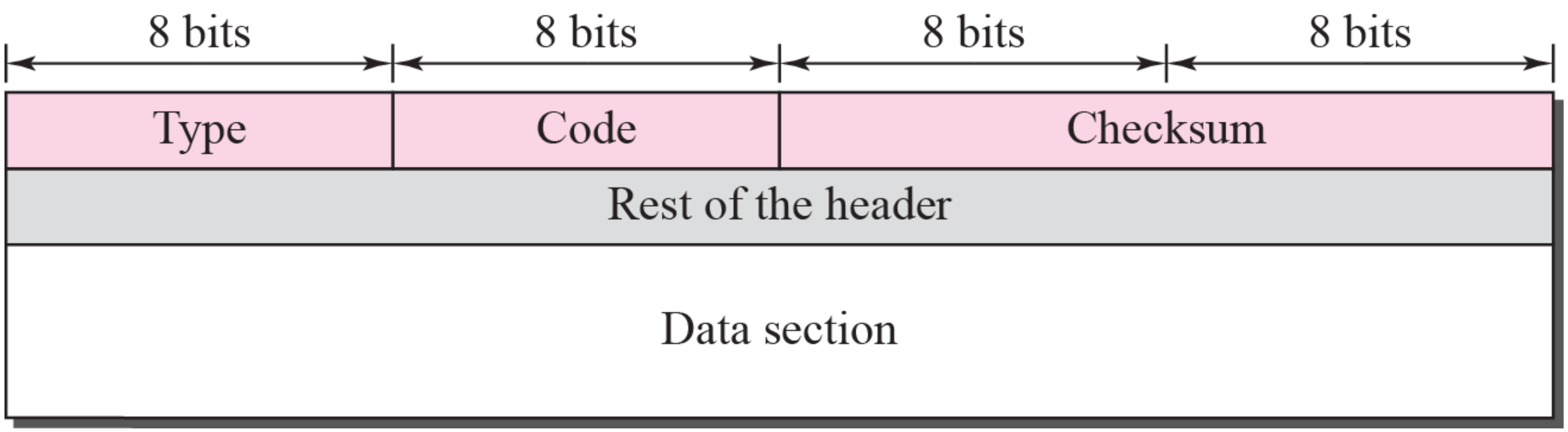
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

Type	Code	Checksum	
0	1	2	3
Pointer	Unused		
4	5	6	7
Internet Header + 8 bytes of Original Data Datagram			
8			

Type	Code	Meaning
0	0	Echo Reply
3	0	Net Unreachable
	1	Host Unreachable
	2	Protocol Unreachable
	3	Port Unreachable
	4	Frag needed and DF set
	5	Source route failed
	6	Dest network unknown
	7	Dest host unknown
	8	Source host isolated
	9	Network admin prohibited
	10	Host admin prohibited
	11	Network unreachable for TOS
	12	Host unreachable for TOS
	13	Communication admin prohibited
4	0	Source Quench (Slow down/Shut up)

Type	Code	Meaning
5	0	Redirect datagram for the network
	1	Redirect datagram for the host
	2	Redirect datagram for the TOS & Network
	3	Redirect datagram for the TOS & Host
8	0	Echo
9	0	Router advertisement
10	0	Router selection
11	0	Time To Live exceeded in transit
	1	Fragment reassemble time exceeded
12	0	Pointer indicates the error (Parameter Problem)
	1	Missing a required option (Parameter Problem)
	2	Bad length (Parameter Problem)
13	0	Time Stamp
14	0	Time Stamp Reply
15	0	Information Request
16	0	Informaiton Reply
17	0	Address Mask Request
18	0	Address Mask Reply
30	0	Traceroute (Tracert)

Figure 9.3 *General format of ICMP messages*



Note

ICMP always reports error messages to the original source.

Figure 9.4 *Error-reporting messages*

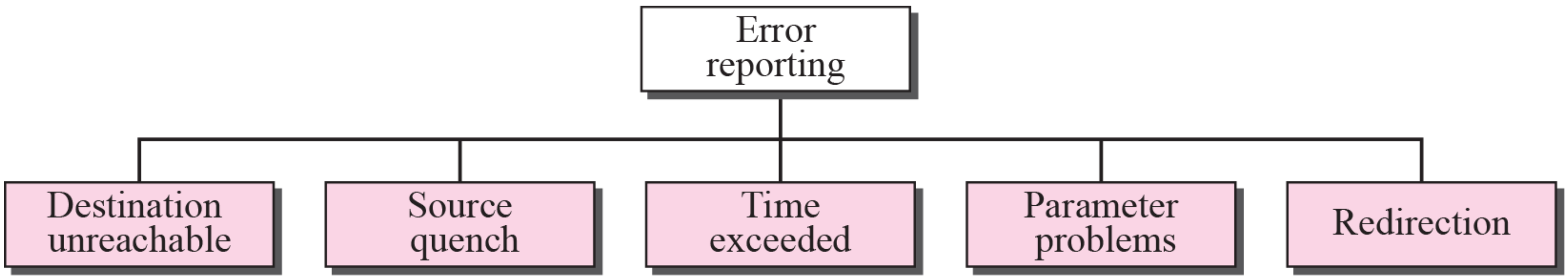


Figure 9.5 *Contents of data field for the error message*

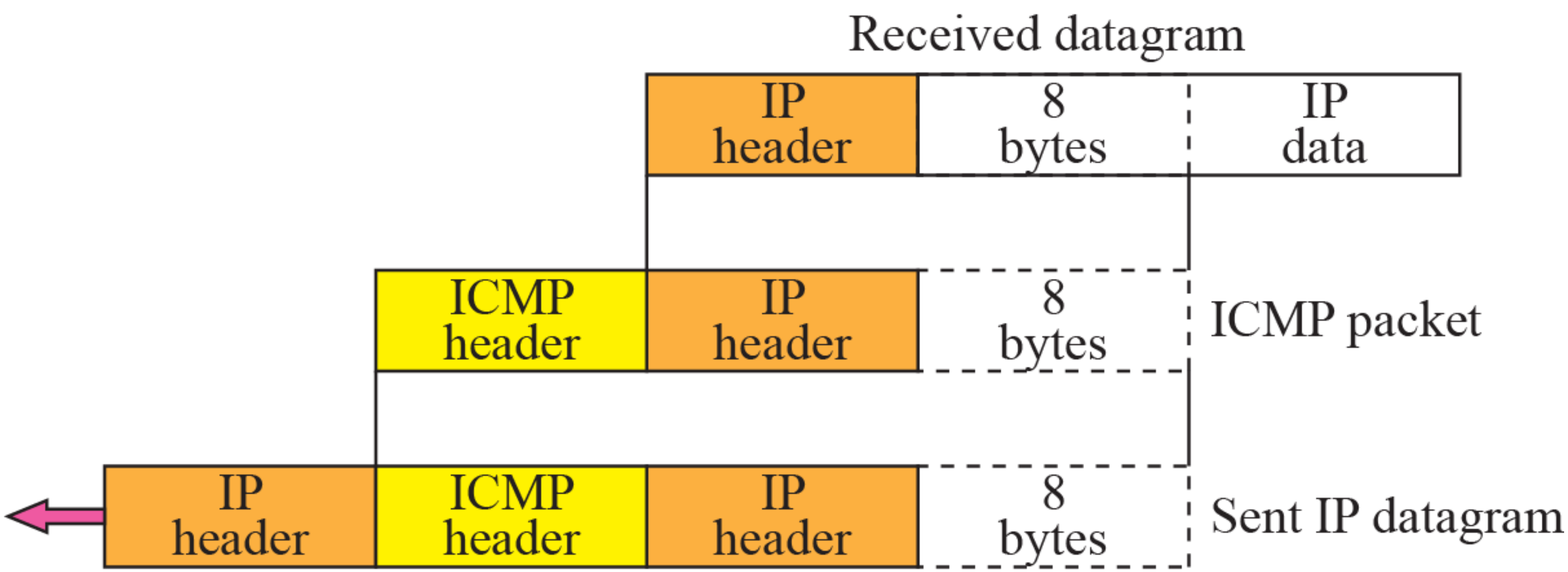


Figure 9.6 *Destination-unreachable format*

Type: 3	Code: 0 to 15	Checksum
Unused (All 0s)		
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

Note

Destination-unreachable messages with codes 2 or 3 can be created only by the destination host.

Other destination-unreachable messages can be created only by routers.

Note

A router cannot detect all problems that prevent the delivery of a packet.

Note

There is no flow-control or congestion-control mechanism in the IP protocol.

Figure 9.7 *Source-quench format*

Type: 4	Code: 0	Checksum
Unused (All 0s)		
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

Note

A source-quench message informs the source that a datagram has been discarded due to congestion in a router or the destination host.

The source must slow down the sending of datagrams until the congestion is relieved.

Note

One source-quench message is sent for each datagram that is discarded due to congestion.

Note

Whenever a router decrements a datagram with a time-to-live value to zero, it discards the datagram and sends a time-exceeded message to the original source.

Note

When the final destination does not receive all of the fragments in a set time, it discards the received fragments and sends a time-exceeded message to the original source.

Figure 9.8 *Time-exceeded message format*

Type: 11	Code: 0 or 1	Checksum
Unused (All 0s)		
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

Note

In a time-exceeded message, code 0 is used only by routers to show that the value of the time-to-live field is zero.

Code 1 is used only by the destination host to show that not all of the fragments have arrived within a set time.

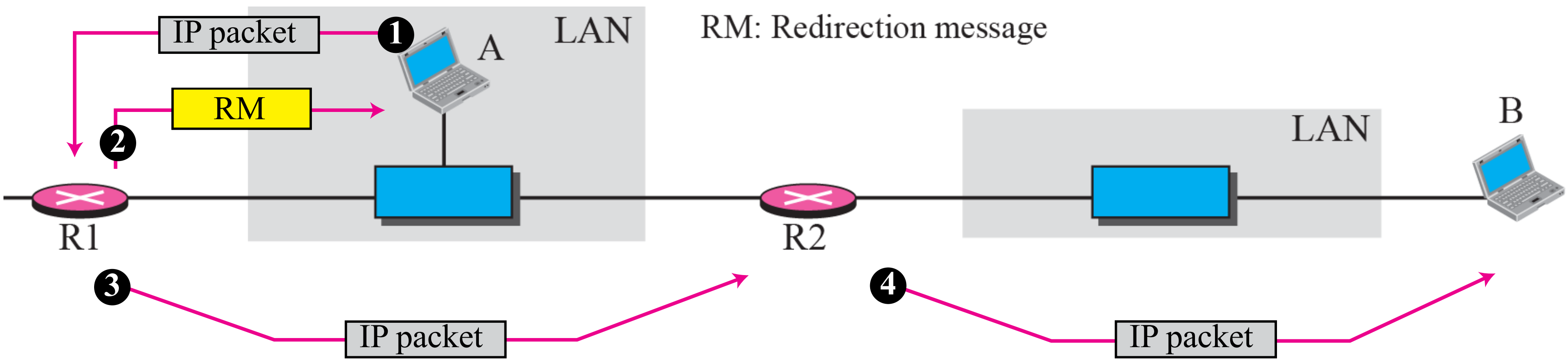
Note

A parameter-problem message can be created by a router or the destination host.

Figure 9.9 *Parameter-problem message format*

Type: 12	Code: 0 or 1	Checksum
Pointer	Unused (All 0s)	
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

Figure 9.10 *Redirection concept*



Note

A host usually starts with a small routing table that is gradually augmented and updated.

One of the tools to accomplish this is the redirection message.

Figure 9.11 *Redirection message format*

Type: 5	Code: 0 to 3	Checksum
IP address of the target router		
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

Note

A redirection message is sent from a router to a host on the same local network.

Note

An echo-request message can be sent by a host or router.

An echo-reply message is sent by the host or router that receives an echo-request message.

Note

Echo-request and echo-reply messages can be used by network managers to check the operation of the IP protocol.

Note

Echo-request and echo-reply messages can test the reachability of a host.

This is usually done by invoking the ping command.

Figure 9.12 *Echo-request and echo-reply message*

Type 8: Echo request
Type 0: Echo reply

<table>
<tr><td>Type: 8 or 0</td><td>Code: 0</td><td colspan="2">Checksum</td></tr>
<tr><td colspan="2">Identifier</td><td colspan="2">Sequence number</td></tr>
<tr><td colspan="4">Optional data
Sent by the request message; repeated by the reply message</td></tr>
</table>

Figure 9.13 *Timestamp-request and timestamp-reply message format*

Type 13: request
Type 14: reply

<table>
<tr><td>Type: 13 or 14</td><td>Code: 0</td><td colspan="2">Checksum</td></tr>
<tr><td colspan="2">Identifier</td><td colspan="2">Sequence number</td></tr>
<tr><td colspan="4">Original timestamp</td></tr>
<tr><td colspan="4">Receive timestamp</td></tr>
<tr><td colspan="4">Transmit timestamp</td></tr>
</table>

Note

Timestamp-request and timestamp-reply messages can be used to calculate the round-trip time between a source and a destination machine even if their clocks are not synchronized.

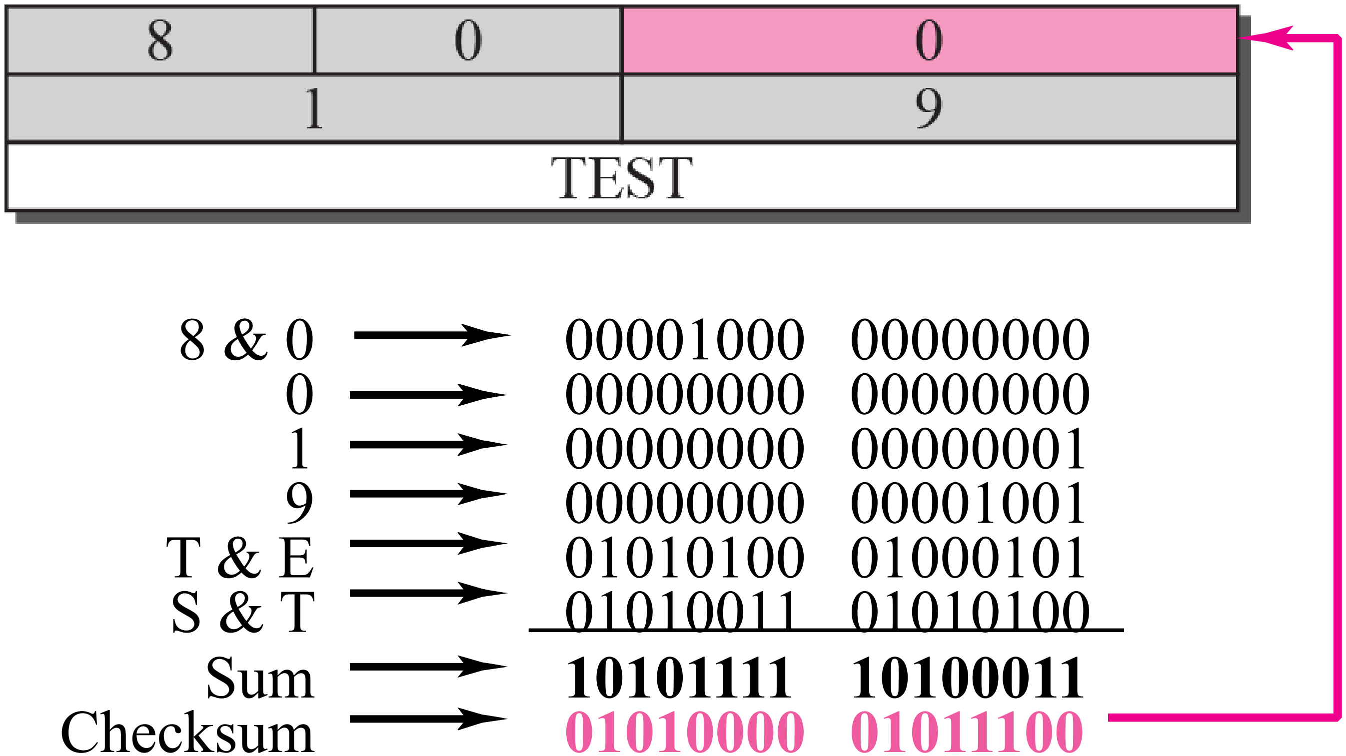
Note

The timestamp-request and timestamp-reply messages can be used to synchronize two clocks in two machines if the exact one-way time duration is known.

Example 9.1

Figure 9.14 shows an example of checksum calculation for a simple echo-request message (see Figure 9.12). We randomly chose the identifier to be 1 and the sequence number to be 9. The message is divided into 16-bit (2-byte) words. The words are added together and the sum is complemented. Now the sender can put this value in the checksum field.

Figure 9.14 *Example of checksum calculation*



9-3 DEBUGGING TOOLS

There are several tools that can be used in the Internet for debugging. We can find if a host or router is alive and running. We can trace the route of a packet. We introduce two tools that use ICMP for debugging: ping and traceroute. We will introduce more tools in future chapters after we have discussed the corresponding protocols.

Topics Discussed in the Section

- ✓ Ping
- ✓ Traceroute

Example 9.2

We use the ping program to test the server fhda.edu. The result is shown below:

```
$ ping fhda.edu
PING fhda.edu (153.18.8.1)    56 (84)  bytes of data.
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=0     ttl=62     time=1.91 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=1     ttl=62     time=2.04 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=2     ttl=62     time=1.90 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=3     ttl=62     time=1.97 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=4     ttl=62     time=1.93 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=5     ttl=62     time=2.00 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=6     ttl=62     time=1.94 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=7     ttl=62     time=1.94 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=8     ttl=62     time=1.97 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=9     ttl=62     time=1.89 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=10    ttl=62     time=1.98 ms

--- fhda.edu ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10103 ms
rtt min/avg/max = 1.899/1.955/2.041 ms
```

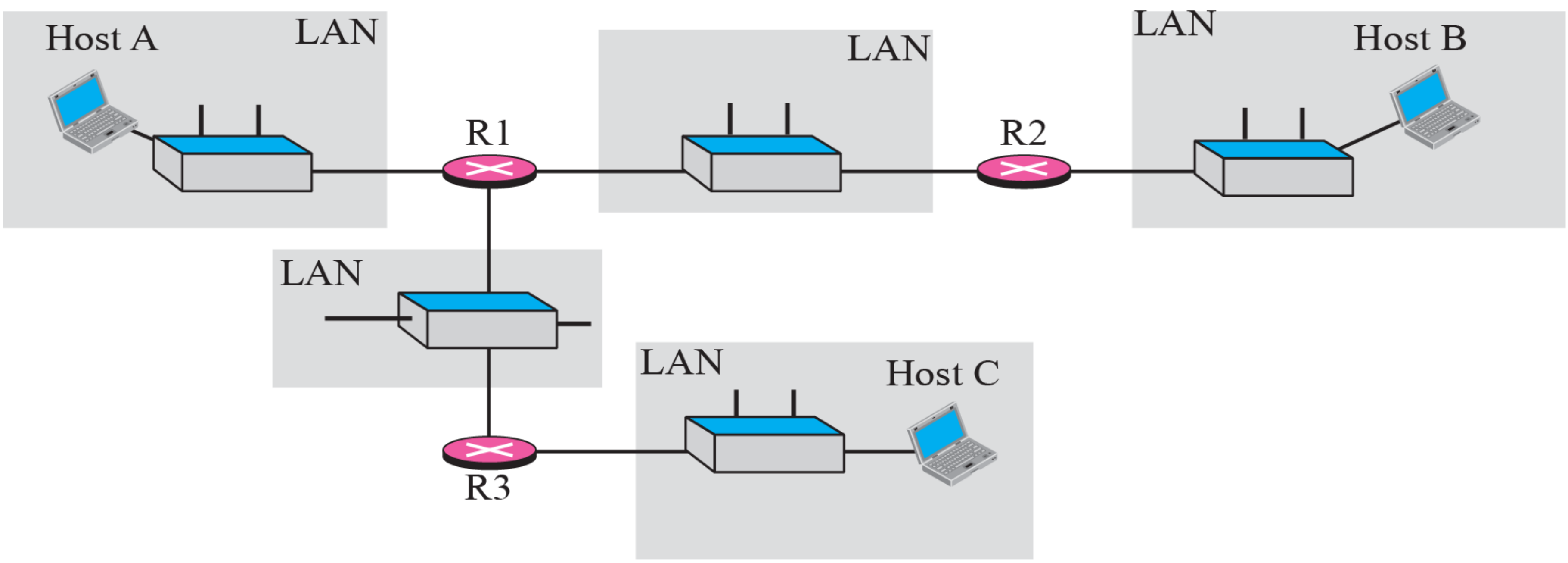
Example 9.3

For the second example, we want to know if the adelphia.net mail server is alive and running. The result is shown below: Note that in this case, we sent 14 packets, but only 13 have been returned. We may have interrupted the program before the last packet, with sequence number 13, was returned.

```
$ ping mail.adelphia.net
PING mail.adelphia.net (68.168.78.100) 56(84) bytes of data.
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=0   ttl=48   time=85.4 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=1   ttl=48   time=84.6 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=2   ttl=48   time=84.9 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=3   ttl=48   time=84.3 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=4   ttl=48   time=84.5 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=5   ttl=48   time=84.7 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=6   ttl=48   time=84.6 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=7   ttl=48   time=84.7 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=8   ttl=48   time=84.4 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=9   ttl=48   time=84.2 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=10  ttl=48   time=84.9 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=11  ttl=48   time=84.6 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=12  ttl=48   time=84.5 ms

--- mail.adelphia.net ping statistics ---
14 packets transmitted, 13 received, 7% packet loss, time 13129 ms
rtt min/avg/max/mdev = 84.207/84.694/85.469
```

Figure 9.15 *The traceroute program operation*



Example 9.4

We use the traceroute program to find the route from the computer voyager.deanza.edu to the server fhda.edu. The following shows the result.

```
$ traceroute fhda.edu
traceroute to fhda.edu      (153.18.8.1), 30 hops max, 38 byte packets
1 Dcore.fhda.edu            (153.18.31.25)   0.995 ms   0.899 ms   0.878 ms
2 Dbackup.fhda.edu          (153.18.251.4)   1.039 ms   1.064 ms   1.083 ms
3 tiptoe.fhda.edu           (153.18.8.1)     1.797 ms   1.642 ms   1.757 ms
```

Example 9.5

In this example, we trace a longer route, the route to xerox.com. The following is a partial listing.

```
$ traceroute xerox.com
```

traceroute to xerox.com (13.1.64.93), 30 hops max, 38 byte packets

1	Dcore.fhda.edu	(153.18.31.254)	0.622 ms	0.891 ms	0.875 ms
2	Ddmz.fhda.edu	(153.18.251.40)	2.132 ms	2.266 ms	2.094 ms
3	Cinic.fhda.edu	(153.18.253.126)	2.110 ms	2.145 ms	1.763 ms
4	cenic.net	(137.164.32.140)	3.069 ms	2.875 ms	2.930 ms
5	cenic.net	(137.164.22.31)	4.205 ms	4.870 ms	4.197 ms
6	cenic.net	(137.164.22.167)	4.250 ms	4.159 ms	4.078 ms
7	cogentco.com	(38.112.6.225)	5.062 ms	4.825 ms	5.020 ms
8	cogentco.com	(66.28.4.69)	6.070 ms	6.207 ms	5.653 ms
9	cogentco.com	(66.28.4.94)	6.070 ms	5.928 ms	5.499 ms

Example 9.6

An interesting point is that a host can send a traceroute packet to itself. This can be done by specifying the host as the destination. The packet goes to the loopback address as we expect.

```
$ traceroute voyager.deanza.edu
traceroute to voyager.deanza.edu   (127.0.0.1), 30 hops max, 38 byte packets
1  voyager          (127.0.0.1)          0.178 ms          0.086 ms          0.055 ms
```

Example 9.7

Finally, we use the traceroute program to find the route between fhda.edu and mhhe.com (McGraw-Hill server). We notice that we cannot find the whole route. When traceroute does not receive a response within 5 seconds, it prints an asterisk to signify a problem (not the case in this example), and then tries the next hop.

```
$ traceroute mhhe.com
traceroute to mhhe.com (198.45.24.104), 30 hops max, 38 byte packets

1    Dcore.fhda.edu       (153.18.31.254)      1.025 ms     0.892 ms     0.880 ms
2    Ddmz.fhda.edu        (153.18.251.40)      2.141 ms     2.159 ms     2.103 ms
3    Cinic.fhda.edu       (153.18.253.126)     2.159 ms     2.050 ms     1.992 ms
4    cenic.net            (137.164.32.140)     3.220 ms     2.929 ms     2.943 ms
5    cenic.net            (137.164.22.59)      3.217 ms     2.998 ms     2.755 ms
6    SanJose1.net         (209.247.159.109)    10.653 ms    10.639 ms    10.618 ms
7    SanJose2.net         (64.159.2.1)         10.804 ms    10.798 ms    10.634 ms
8    Denver1.Level3.net   (64.159.1.114)       43.404 ms    43.367 ms    43.414 ms
9    Denver2.Level3.net   (4.68.112.162)       43.533 ms    43.290 ms    43.347 ms
10   unknown              (64.156.40.134)      55.509 ms    55.462 ms    55.647 ms
11   mcleodusa1.net       (64.198.100.2)       60.961 ms    55.681 ms    55.461 ms
12   mcleodusa2.net       (64.198.101.202)     55.692 ms    55.617 ms    55.505 ms
13   mcleodusa3.net       (64.198.101.142)     56.059 ms    55.623 ms    56.333 ms
14   mcleodusa4.net       (209.253.101.178)    297.199 ms   192.790 ms   250.594 ms
15   eppg.com             (198.45.24.246)      71.213 ms    70.536 ms    70.663 ms
16   ...                  ...                  ...          ...          ...
```

9-4 ICMP PACKAGE

To give an idea of how ICMP can handle the sending and receiving of ICMP messages, we present our version of an ICMP package made of two modules: an input module and an output module.

Topics Discussed in the Section

- ✓ Input Module
- ✓ Output Module

Figure 9.16 *ICMP package*

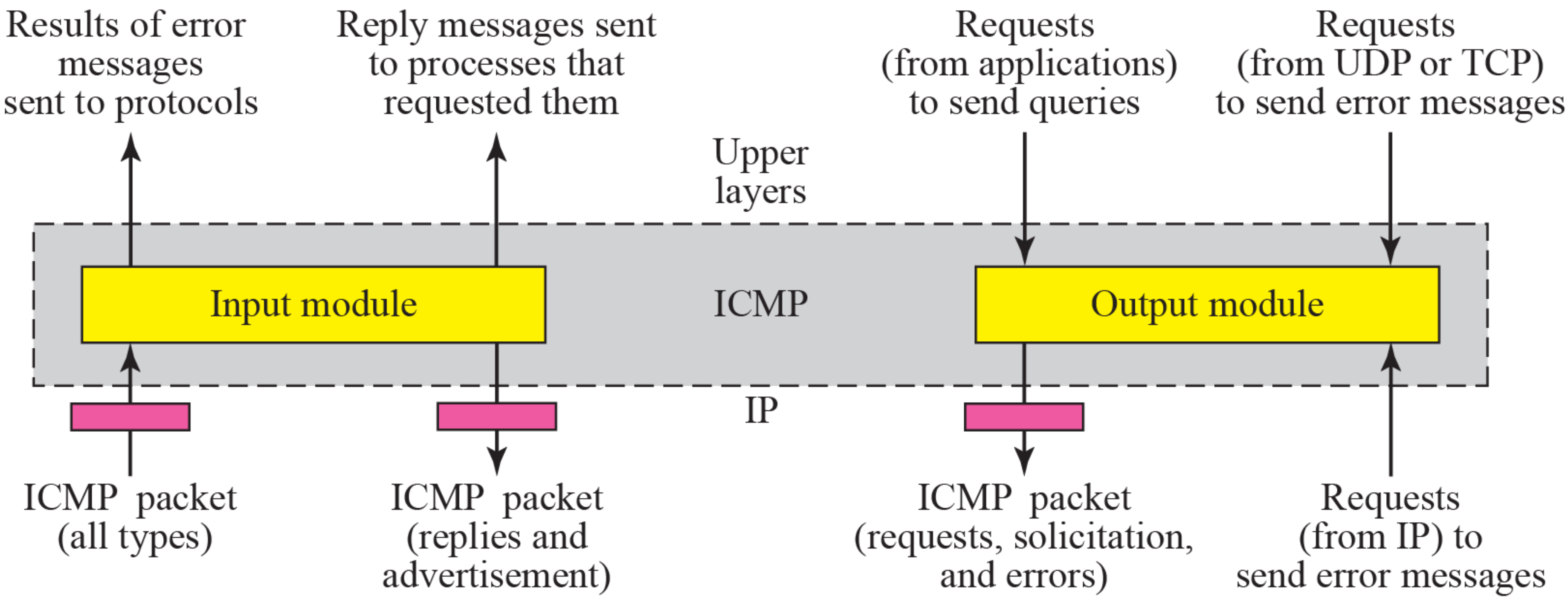


Table 9.2 *Input Module*

```
ICMP_Input_module (ICMP_Packet)
{
    If (the type is a request)
    {
        Create a reply
        Send the reply
    }
    If (the type defines a redirection)
    {
        Modify the routing table
    }
    If (the type defines other error messages)
    {
        Inform the appropriate source protocol
    }
    Return
}
```

Table 9.3 *Output Module*

```
ICMP_Output_Module (demand)
{
    If (the demand defines an error message)
    {
        If (demand comes from IP AND is forbidden)
        {
            Return
        }
        If (demand is a valid redirection message)
        {
            Return
        }
        Create an error message
    If (demand defines a request)
    {
        Create a request message
    }
    Send the message
    Return
}
```